After Making Relationships Visible: Historical Judgement in Visualisation

After this week's readings, I am less concerned with whether historians can use digital tools than with what more these tools allow us to see. In my reading notes, I tend to view network analysis as a process of gradual “narrowing”: first broadening the search for connections, then selecting relationships around a research question. Visualisation should likewise help readers understand how research reaches judgements, rather than merely make results look attractive. Whether a tool is worth using depends on its ability to help pose questions, test interpretations or communicate evidence. Historians have long studied networks; digital technology facilitates collecting, comparing and measuring data. Using “network” as a metaphor alone does not analyse relational structure.

Correspondence research illustrates this value. Letters are themselves a medium connecting people; separately coding sending, receiving, forwarding and introductions links actions scattered across sources. Bourke's discussion of women's history suggests that network analysis can reveal someone's “hidden” role as intermediary or supporter. Yet it is not inherently better suited to women's history than close reading. Its value depends on whether relational data supplement narratives centred on prominent individuals. This approach also applies to knowledge transmission, commerce and kinship. Questions involving wording, silence or emotion still require returning to individual texts.[1]

I initially wondered whether studying Sidney women could begin by designating them as hubs before collecting letters. Starting with women helps uncover overlooked material. However, collecting only their connections and then treating their central position as proof of importance is circular. Numbers of contacts, connections to important people and links between groups concern degree, eigenvector and betweenness centrality respectively; they cannot all be labelled “influence”. High scores primarily indicate positions within a particular dataset. Actual roles still require interpreting the letters.[1]

I therefore think narrowing must follow clarification of the dataset's boundaries. Relationships that do not fit expectations should not be removed for visual neatness. In my research, I would try first to define the period and sources, recording name variants, relationship types, dates, provenance and grounds for interpretation. I would distinguish directly documented relationships from inference and retain uncertain identities. Sender–recipient metadata suit the study of communication structures, but cannot adequately explain

how women are discussed in letters. Name changes and appearances only in others' biographies may cause identification rules to exclude women. Missing letters do not mean absent relationships; enemies may correspond frequently, so connection counts cannot directly measure intimacy. Researchers need not enter everything themselves, but should help define categories, resolve ambiguities and conduct spot checks. Otherwise, efficiency merely brings bias into the graph more quickly.[1]

Medici's Sidney network shows how such judgements shape presentation. Colours distinguish correspondence, news, introductions and other relationships more meaningfully than uniform lines. Yet names overlap in dense areas, while combining samples from 1565–1566, 1590 and 1595 can make successive relationships appear simultaneous. I would retain the overview as an index, offer views filtered by period and relationship type, and allow readers to inspect each connection's evidence. “Narrowing” should make selection visible, rather than present a reduced graph as the complete network.[2]

Theibault's emphasis on information density and transparency reminds me that simplification need not weaken arguments; what matters is what gets simplified. Minard's map of Napoleon's campaign connects routes, line widths representing troop numbers and temperatures during the retreat. Showing movement and losses together gives this static image considerable narrative power. However, visually linking temperatures and declining numbers cannot establish that cold caused every loss. If using it in my argument, I would add a caption explaining that declining troop numbers do not simply mean deaths, and clarify which causes of loss the map cannot distinguish. This would prevent a clear image from suggesting an overly strong causal explanation.[3]

Salmi's Voyant example reveals another limit. Highlighting “history” and “digital”, the word cloud quickly indicates the text's topic, but cannot establish the historical arguments these words convey. The adjacent frequency curves show distribution across the text, not change over historical time. I would use the cloud as a search starting point, adding context and comparable frequency tables. If a few sentences suffice, no extra chart is needed. Salmi also acknowledges that the example only touches the surface. We should therefore judge whether a visualisation fulfils its specific task, rather than always demand new discoveries.[4]

These examples also revise my understanding of “highlighting time”: adding a timeline is insufficient without explaining periodisation, uncertain dates and events' different rhythms. Based on my current understanding of Gephi and Datawrapper, I would use Gephi to explore

relational structures, Datawrapper to compare numerical changes across periods, and close reading to explain anomalies. I would not directly treat distances in a graph's layout as social distances. When a network contains different node types, such as people and texts, I would also check whether the metrics are appropriate. Before comparing periods, I would check surviving source counts to avoid mistaking preservation differences for changing interaction. Comparing results with and without inferred relationships would test whether “central figures” change with filtering rules. This integrates historical judgement into data and visual design, rather than waiting for software to explain the past for me.[1][4]

I still do not think every historical study must use digital methods. Bourke's discussion of funding, learning costs and data work reminds me that resource allocation shapes technical choices. Those prioritised for digitisation become more visible in history; adding a few women does not eliminate archival bias.[1] For public audiences, I would want visualisations to attract attention while disclosing gaps, providing written explanations and versions that can be preserved, and crediting those who compile and correct data. Digital history also extends beyond the internet: classroom presentations and digital museum displays help the public understand history. Web layouts, legends and annotations determine how easily readers find explanations and evidence.[3][4] As I noted, digital history offers new ways to encounter the past, but openness should include readers' right to question images and check evidence.

References and Visualisations

[1] Evan Bourke, “Networking early modern Irish women,” Irish Historical Studies 46, no. 170 (2022): 270–285, especially 273–284.

[2] Catherine Medici, “Visualizing the Sidney Network,” project webpage and “Networks with Sample Years” image, accessed 4 October 2026. Original image

[3] John Theibault, “Visualizations and Historical Arguments,” in Writing History in the Digital Age, ed. Kristen Nawrotzki and Jack Dougherty (2013), 173–185, especially 174–178 and Figure 7.

[4] Hannu Salmi, “Presenting the Past in the Digital Age,” in What is Digital History? (Polity, 2020), supplied bilingual PDF, pp. 2–8 and Figure 3.

Statement: AI tools (ChatGPT) were utilised for language translation. All notes and the manuscript were initially composed in Chinese.